

Lab 1

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Problem 1

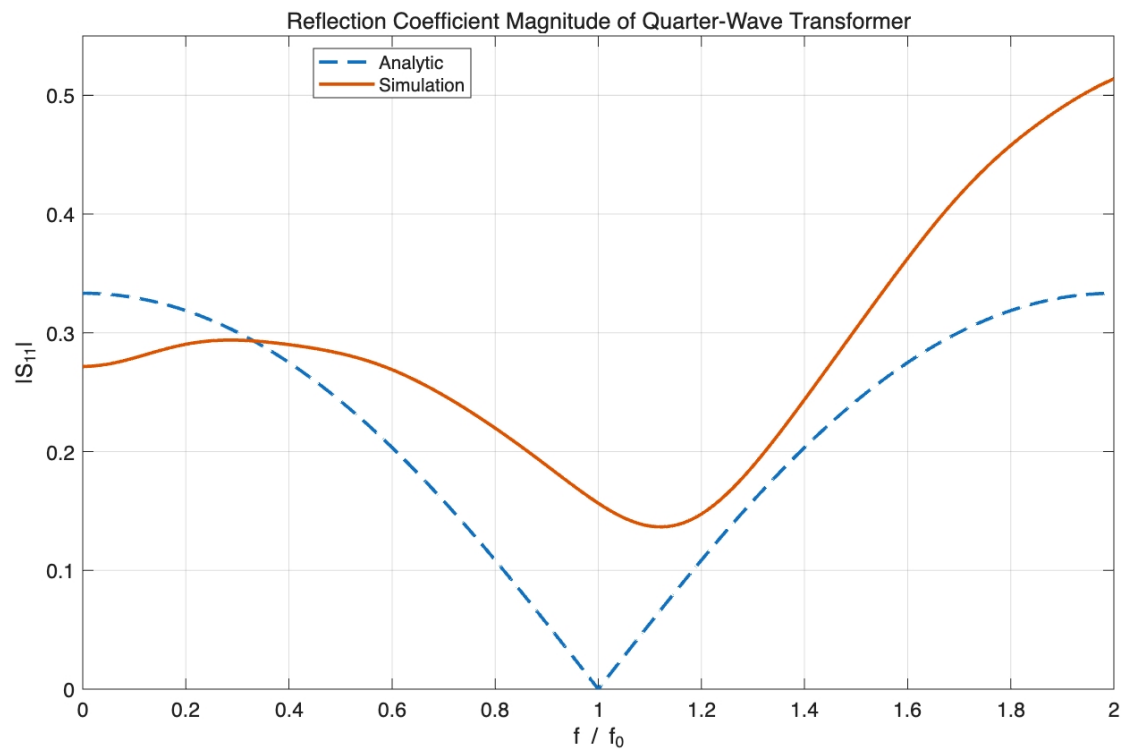
We first calculate the theoretical length and width of the QWT:

Case 1: $Z_C = 70.71\Omega$

Case 2: $Z_C = 100\Omega$

Case 1	W=1.5543mm	L=7.2241mm
Case 2	W=0.6778mm	L=7.4062mm

We run case 1 in CST and import the data into Matlab, then we get a figure as below:



The simulated plot is seemingly different from an ideal analytical QWT model: the minimum does not reach zero and is shifted to a slightly higher frequency, around $f/f_0 \approx 1.1$. This means the implemented microstrip transformer is not an ideal quarter-wave transformer at f_0 .

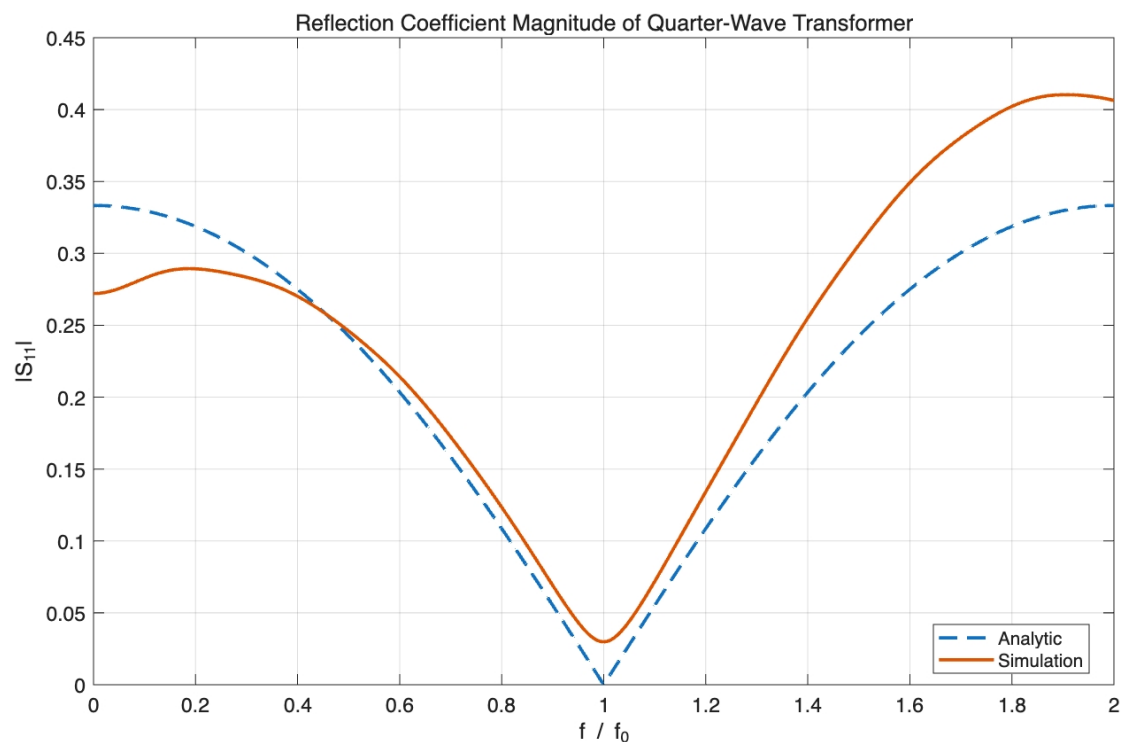
Moreover, the CST curve is asymmetric and becomes much larger than the analytical one at higher frequencies. This shows that the real microstrip structure has additional effects that are not included in the simple transmission-line model, such as dispersion, finite conductor thickness, non-ideal port/load implementation and so on.

So in order to make the characteristic impedance more closer to the designed value as well as to have the minimum S_{11} exactly at 5.8 GHz, we tune the parameters as below:

		W(mm)	L(mm)
Case 1	original	1.5543	7.2241
	tuned	1.48	8.25
Case 2	original	0.6778	7.4062
	tuned	0.565	7.47

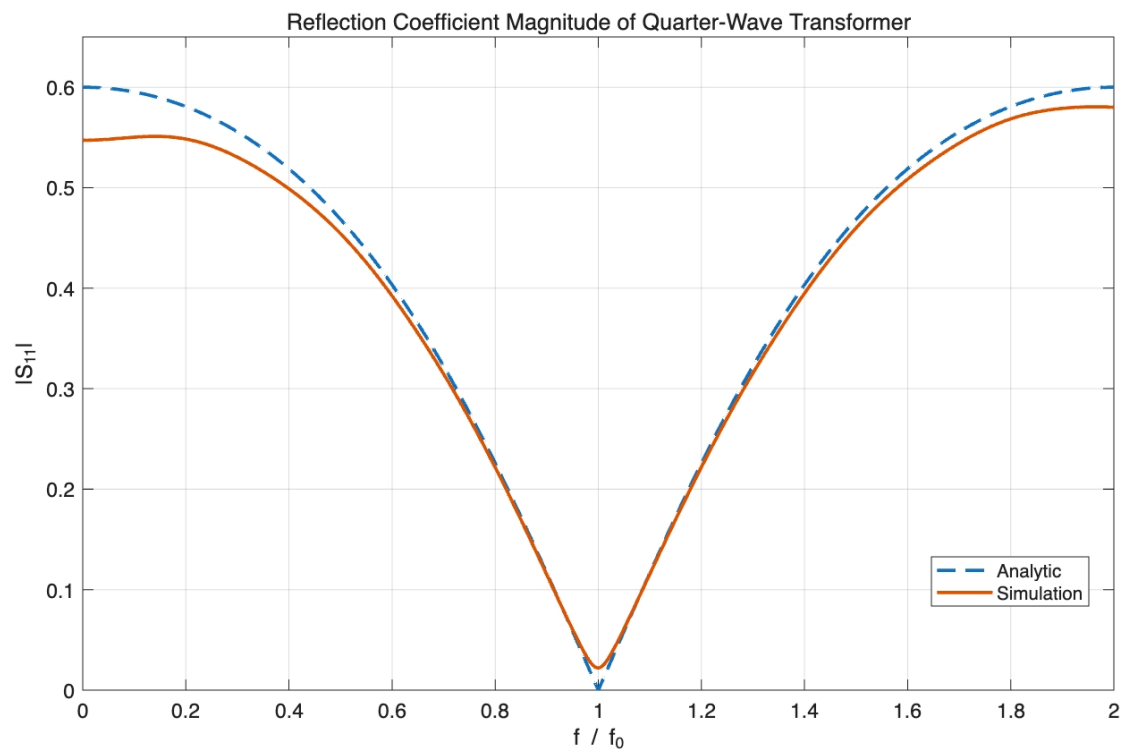
After tuning we use Matlab to get the figures:

Case 1:



For **Case 1**, after tuning the width and length, the minimum reflection moves much closer to 5.8 GHz and the overall response becomes closer to the theoretical QWT behavior. However, some discrepancy still remains, especially at higher normalized frequencies, due to non-ideal microstrip effects.

Case 2:



For **Case 2**, the tuned result matches the analytical curve very well. This indicates that the tuned geometry reproduces the intended QWT behavior much better than the initial design.

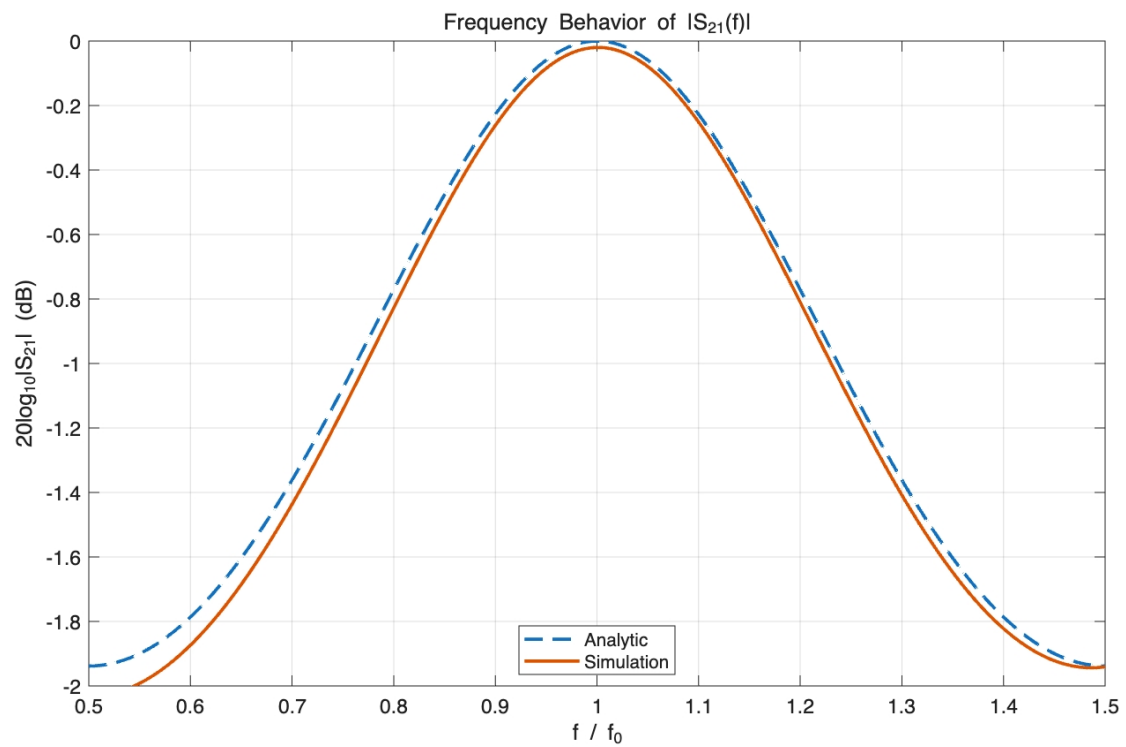
Problem 2

As is same with **Problem 1**, when we use theoretical value to run the CST solver we obtain characteristic impedance and minimum- S_{11} -frequency different from our ideal value, so we tune the parameters:

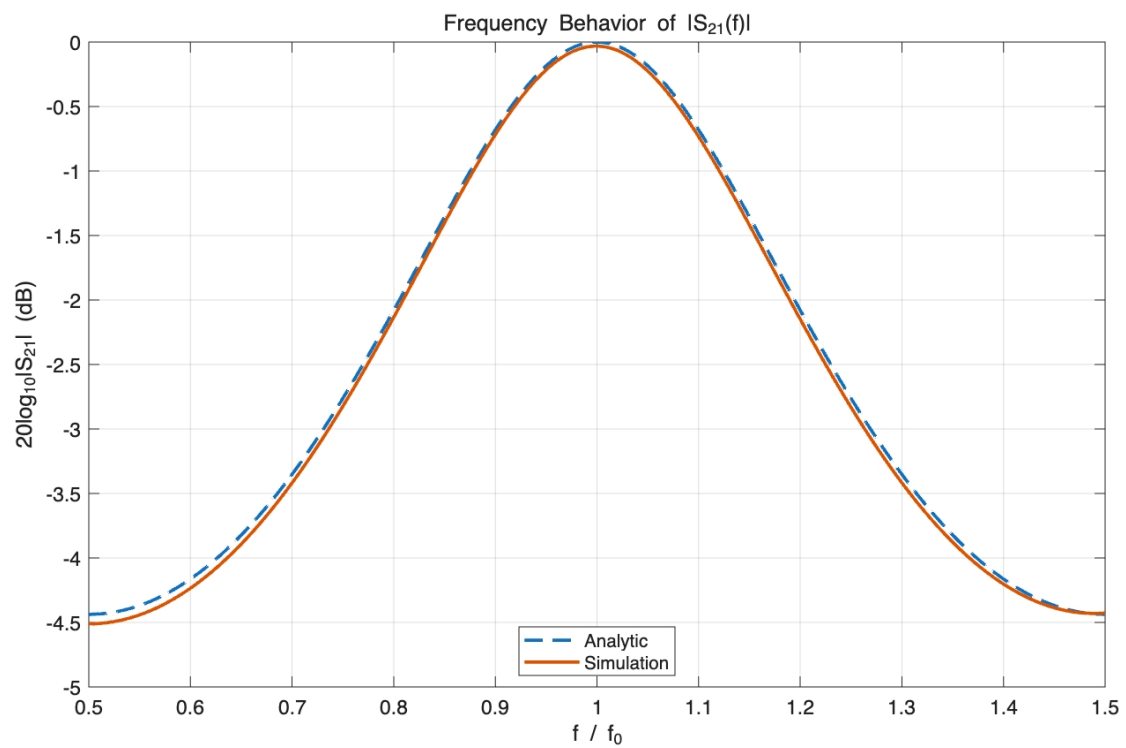
		W(mm)	L(mm)
Case1 $Z_c=100\Omega$	original	0.6778	14.8123
	tuned	0.55	14.55
Case 2 $Z_c=150\Omega$	original	0.1703	15.2148
	tuned	0.0925	15.1
Case 3 $Z_c=10\Omega$	original	24.1757	12.8970
	tuned	24.7	12.5

We then run the solver in CST and get the figures from Matlab:

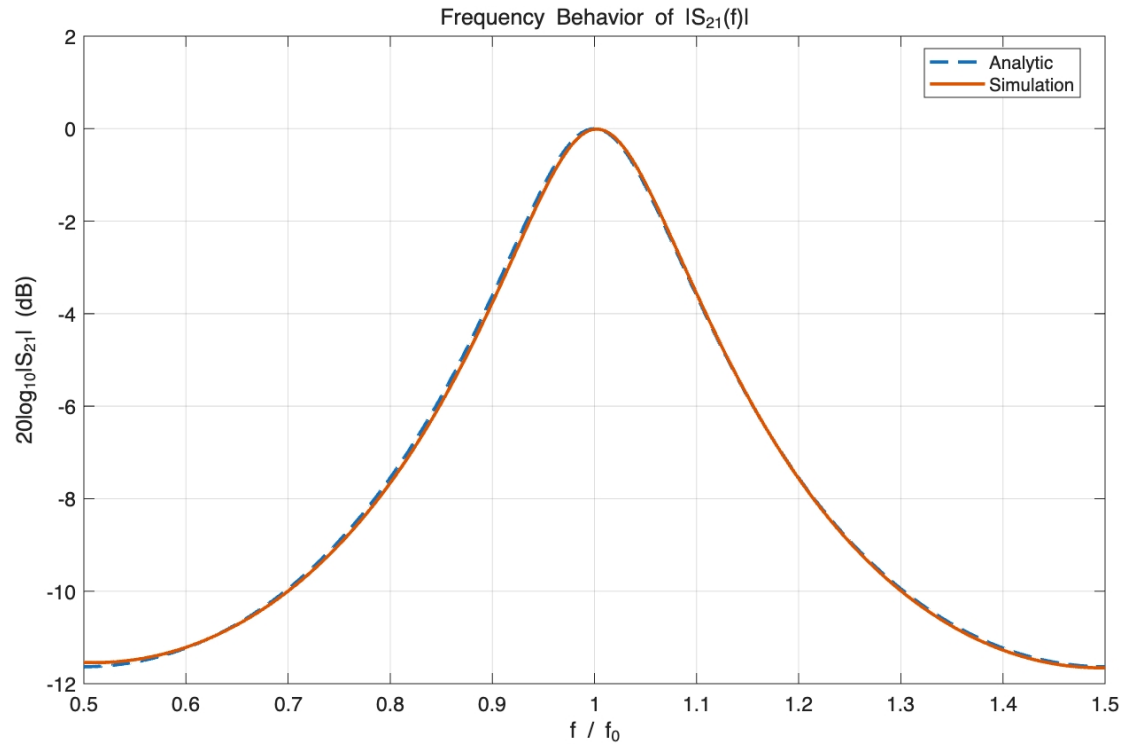
Case 1:



Case 2:



Case 3:



For all the three cases, The simulated curve almost overlaps the analytical one over the whole frequency range, which means the tuned dimensions reproduce the expected behavior of the line very accurately.

In conclusion, the tuning process is effective because the dimensions obtained from closed-form formulas are only approximate. By adjusting the width and length, the real CST model can better satisfy both the target characteristic impedance and the quarter-wave condition at the design frequency.